A 3.2.2 Worksheet

Worksheet: Entity Relationship Diagrams (ERDs)

1. Relationship Identification

For each scenario, identify the entities and the type of relationship (one-to-one, one-to-many, many-to-many). For each relationship, also specify the cardinality and modality from both entity perspectives.

a. A student and their student ID.
* Entities:
* Relationship Type:
* Cardinality:
* Student to Student ID:
* Student ID to Student:
* Modality:
* Student to Student ID:
* Student ID to Student:

b. An author and their books.
* Entities:
* Relationship Type:
* Cardinality:
* Author to Book:
* Book to Author:
* Modality:
* Author to Book:
* Book to Author:

c. Students and classes.
* Entities:
* Relationship Type:
* Cardinality:
* Student to Class:
* Class to Student:
* Modality:
* Student to Class:
* Class to Student:

d. A department and its employees.
* Entities:
* Relationship Type:
* Cardinality:
* Department to Employee:
* Employee to Department:
* Modality:
* Department to Employee:
* Employee to Department:

e. A product and its reviews.
* Entities:
* Relationship Type:
* Cardinality:
* Product to Review:
* Review to Product:
* Modality:
* Product to Review:
* Review to Product:

2. Cardinality and Modality

Determine the cardinality and modality for the following relationships. Explain your reasoning in detail.

a. A customer places one or more orders.
* Cardinality:
* Customer to Order:
* Order to Customer:
* Modality:
* Customer to Order:

* Order to Customer:
* Reasoning:

b. A book may or may not be assigned to a library member.
* Cardinality:
* Book to Member:
* Member to Book:
* Modality:
* Book to Member:
* Member to Book:
* Reasoning:

c. A player participates in one team.
* Cardinality:
* Player to Team:
* Team to Player:
* Modality:
* Player to Team:
* Team to Player:
* Reasoning:

d. A song is on a playlist.
* Cardinality:
* Song to Playlist:
* Playlist to Song:
* Modality:
* Song to Playlist:
* Playlist to Song:
* Reasoning:

3. ERD Notation

a. Draw an ERD using Chen notation for a library system, including entities like Books, Authors, and Members.

b. Draw an ERD using Crow's Foot notation for a university system, including entities like Students, Courses, and Professors.